



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 • STANDARD 6.NOS.2

# Divide Multi-Digit Numbers Using an Algorithm

Lesson 2-6 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

**Language Objective:** I can explain my division using the words dividend, divisor, quotient, and remainder.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Algorithm

Dividend

Divisor

Quotient

Remainder

Divide Multi-Digit Numbers

Long division

Bring down



Tap any word to see its meaning and a picture.

1

A step-by-step set of rules to solve a problem is called an

2

In  $84 \div 7$ , the total amount 84 being divided is the

3

In  $84 \div 7$ , the number 7 that you divide by is the

4

The answer to a division problem is called the

5

When a number does not divide evenly, the amount left over is the

6 Long division repeats four steps: divide, multiply, subtract,

.

7 The method that repeats divide, multiply, subtract, and bring down is called

.

8 After you subtract in long division, the next step is to

the next digit.

## Write About the Math

**How does multi-digit division help the school figure out how many pencils each classroom gets?**

*¿Cómo ayuda la división de varios dígitos a la escuela a averiguar cuántos lápices recibe cada salón?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Algorithm**  
*Algoritmo*

☐ **Dividend**  
*Dividendo*

☐ **Divisor**  
*Divisor*

☐ **Quotient**  
*Cociente*

☐ **Remainder**  
*Residuo*

**Model:** *Estimating with friendly numbers checks whether the quotient is reasonable.* — Students estimate using  $1,200 \div 12 = 100$  (or similar) to predict roughly 100+ bars per section, identifying 1,344 as the dividend and 12 as the divisor.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Each classroom gets \_\_\_\_ pencils because  $1,680 \div 24 =$  \_\_\_\_.**
- **My answer is \_\_\_\_ because \_\_\_\_.**  
*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My evidence is \_\_\_\_, so \_\_\_\_.**  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Algorithm
2. **Notes 2:** Dividend
3. **Notes 3:** Divisor
4. **Notes 4:** Quotient
5. **Notes 5:** Remainder
6. **Notes 6:** Divide Multi-Digit Numbers
7. **Notes 7:** Long division
8. **Notes 8:** Bring down
9. **Watch for this mistake:** *A common mistake in Divide Multi-Digit Numbers is skipping a placeholder zero in the quotient when a digit doesn't divide evenly, which shifts every digit after it out of place. For example, dividing  $3,216 \div 3$ :  $3 \div 3 = 1$ , but the next digit 2 doesn't divide by 3, so a 0 must be written before continuing — bring down the 1 to make  $21 \div 3 = 7$ , then bring down 6  $\div 3 = 2$ , giving the correct quotient 1,072. A student who skips writing that 0 and jumps straight to  $21 \div 3 = 7$  ends up with 172 instead of 1,072 — an answer off by a factor of ten. Always check: does a digit in your quotient go missing every time you 'can't divide'? If so, you dropped a placeholder zero.*
10. **Try It 1:** C.  $704 - 7 \times 704 = 4,928$  exactly, so the quotient is 704 with no remainder.
11. **Try It 2:** B.  $671 \text{ R } 6 - 9 \times 671 = 6,039$ , and  $6,045 - 6,039 = 6$ , so  $6,045 \div 9 = 671 \text{ R } 6$ .